Surrogate-based stability of Pega ADM explanations

Thesis progress + L5B15 Health Check findings

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Agenda

1. **Where the thesis stands**, stability of feature attributions
2. **Bonus: L5B15 Health Check findings**, what Pega's own tool reveals
3. **Forward look**, predictors that could lift luggage CTR
4. **Next steps & open questions**

Framing

Sections 1 & 4 are the academic deliverable.
Sections 2 & 3 are *hypotheses surfaced by the Health Check review*, directions to validate, not actions to take tomorrow.

PART 1

Where the thesis stands

Three research questions

RQ1 How accurately do SHAP, LIME and Decision Tree surrogates approximate the Pega black-box propensity model?

RQ2 How stable are feature attribution rankings across temporal splits and route subgroups?

RQ3 Does observed instability reflect data drift (δ_d), model churn (δ_m), or explainer sensitivity (δ_e)?

Scope

Luggage upsell, Email/Outbound channel. Primary variant: **L5B15.** Replicated on CLUG, BookingDotCom, Cartrawler.

Why a CatBoost surrogate, and how well it works

Pega ADM uses Naive Bayes internally. We train CatBoost *on* Pega's output propensities.

- **TreeSHAP requires a tree model**, KernelSHAP on NB would $\sim 100\times$ compute and stack its own error
- **Fidelity over mimicry**, flexible surrogate reproduces NB's mapping better than one constrained by conditional independence
- **Native categorical + missing handling**, no encoding choices as confounds

Variant	**R^2**	**Spearman ρ**	**RMSE**	**n**
L5B15	0.927	0.933	0.0026	31,709
CLUG	0.954	0.920	0.0026	16,984
BookingDotCom	0.793	0.949	0.0002	19,866
Cartrawler	0.912	0.615	0.0005	20,645

High fidelity on three of four variants. Cartrawler's low ρ flags propensity-rank degradation despite good R^2.

Instability decomposition (RQ3) in plain language

We decompose how much each **ranking** changes between two splits into three sources:

$$\begin{aligned}
\textbf{DT instability} &= \text{data drift} + \text{model churn} + \text{explainer gap} \\
\textbf{SHAP instability} &\approx \text{data drift} + \text{model churn} \\
\textbf{Explainer gap} &= \text{DT instability} - \text{SHAP instability}
\end{aligned}$$

Data drift
KS statistic on the propensity score distribution (bin-free, non-parametric).

Model churn
Jaccard overlap on the set of `modelVersion` IDs active in each split (low overlap = high churn).

Explainer gap
What DT picks up that SHAP does not, i.e. sensitivity of the explanation method itself.

Evaluation design

Each split is a pair of sub-populations evaluated on the same surrogate.

- **Temporal split**, early vs late window within each variant
- **Route subgroups**, top 4 routes by volume (e.g. ORY→NCE, ORY→OPO, NCE→ORY, TLS→ORY)
- **Culture / language split**, fr-FR vs nl-NL bookings
- **Replication**, L5B15 → CLUG, BookingDotCom, Cartrawler

Why same-window splits are useful

Route and culture splits share a time window, so the active model snapshots are essentially the same: $\delta_m \approx 0$ and δ_d stays small. Any remaining instability is mostly δ_e.

A within-route DT bootstrap ($N = 10$ seeds) gives the noise floor for DT ranking variability, so we can tell “real” between-route disagreement from fitting noise.

Stability results: L5B15, by split

Spearman ρ between top-feature rankings (closer to 1 = more stable):

Split	DT ρ	LIME ρ	SHAP ρ
Temporal (early vs late)	0.855	0.989	0.996
Route: NCE$\rightarrow$ORY vs TLS$\rightarrow$ORY	0.817	0.992	0.987
Route: ORY$\rightarrow$NCE vs ORY$\rightarrow$OPO	**0.510**	0.986	0.994
Route: ORY$\rightarrow$OPO vs NCE$\rightarrow$ORY	**0.584**	0.858	0.990
Culture: fr-FR vs nl-NL	**0.667**	0.949	0.958

Red highlight: DT $\rho < 0.7$, i.e. rankings disagree on more than 30% of pairs.

Pattern

SHAP and LIME are very stable ($\rho > 0.95$ almost everywhere). **DT rankings collapse** on certain route and language splits (ρ down to 0.51), well below the within-route DT bootstrap noise floor (mean $\rho \approx 0.83$ across 10 seeds), so the disagreement is real, not a fluke of how the tree was fit.

Instability decomposition: where does it come from?

Split	Data drift (KS)	Model churn (J_v)	Explainer gap	Dominant
Temporal (early vs late)	0.36	0.00	0.141	external shift
ORY→NCE vs ORY→OPO	0.10	0.48	0.484	**explainer gap**
ORY→OPO vs NCE→ORY	0.09	0.55	0.406	**explainer gap**
NCE→ORY vs TLS→ORY	0.09	0.52	0.171	**explainer gap**
fr-FR vs nl-NL	0.08	0.59	0.291	**explainer gap**

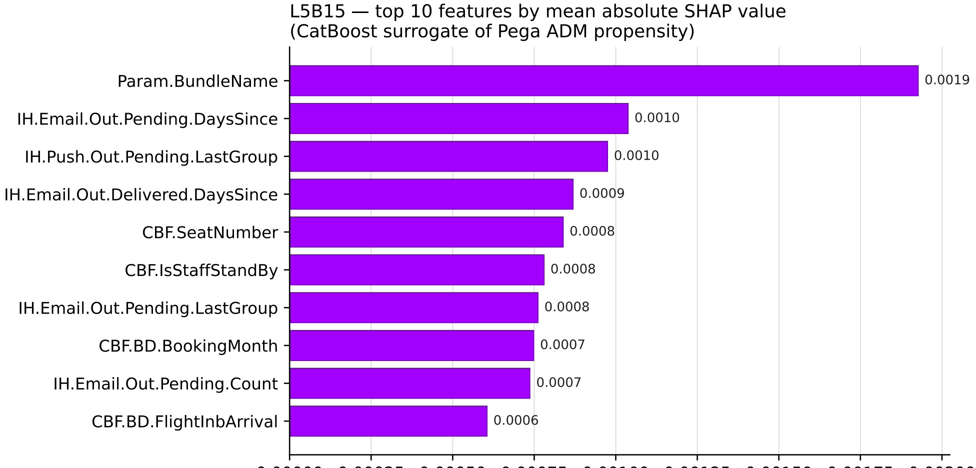
KS in [0, 1]; significant drift if ≥ 0.10. J_v is Jaccard overlap of active `modelVersion` sets; *higher* = *more stable*; significant churn if $J_v \leq 0.10$. Same-window splits sit near $J_v \approx 0.5$ by design, since Pega's online NB issues a fresh `modelVersion` UUID on every micro-update.

Headline finding (RQ3)

On route and culture subgroups, neither the input data nor the active model versions change much, yet DT's top-feature list looks very different across them. SHAP rankings stay almost identical on the same splits, so the difference comes from the explanation method itself, not from the model. **DT can disagree with itself across populations even when the underlying model has not changed.**

What SHAP says are the most important features

Top 10 L5B15 features by mean absolute SHAP value, from the CatBoost surrogate:



PART 2

L5B15 Health Check findings

The Pega ADM Health Check

- Pega's official `pdstools` tool, run on the same Datamart export this thesis uses
- Generic ADM overview *plus* per-model deep-dives
- I ran it specifically against **Sales** / **Luggage** / **L5B15** / **L5B15_EM_T01**
- AUC = **76.56**, responses = 2.86 M, positives = 25,766 **healthy model**

Why this matters here

The Health Check views the model through Pega's own lens (univariate NB performance). Comparing it side-by-side with our surrogate's SHAP/LIME/DT importances is a direct external sanity check on RQ1 and a free source of operational insights.

Where L5B15 sits in the wider ADM picture

Only 36%

of the 944 ADM models in the snapshot are *mature and useful*.

(mature = $\geq$ 200 positives and AUC $\geq$ 0.52)

- Email / Outbound channel: 959K positives, positive-response rate 1.63%, average AUC 69.1
- Luggage group: 70 models (L5B15, L5B20, CLUG, ...)
- 220 models have never been used or never received a positive, dead weight in the system

L5B15: what the NB model actually uses

Of 220+ predictors in the pool, only **24 are Active** in the NB model.
Top 10 by univariate AUC (Pega's own measure, from Predictor Overview table):

#	Predictor	AUC	Note
1	`Param.BundleName`	74.28	action-label proxy **verify**
2	`CustBookedFlight.BookingData.BookingMonth`	72.50	seasonality **verify**
3	`IH.Email.Out.Pending.pxLastOutcomeTime.DaysSince`	67.42	IH recency
4	`IH.Email.Out.Pending.pyHistoricalOutcomeCount`	66.21	IH volume
5	`CustBookedFlight.IsStaffStandBy`	66.12	**data-gap proxy**
6	`IH.Push.Out.Pending.pyHistoricalOutcomeCount`	65.63	cross-channel
7	`IH.Push.Out.Pending.pxLastOutcomeTime.DaysSince`	64.68	
8	`IH.Email.Out.Pending.pxLastGroupID`	64.67	last-group context
9	`IH.Push.Out.Pending.pxLastGroupID`	64.24	
10	`IH.Email.Out.Delivered.pxLastGroupID`	63.98	

Mostly a recency/fatigue machine; customer profile barely contributes.

Pega marks predictors *Inactive* when their correlation with another active predictor exceeds a tunable threshold (default ~ 0.8) or when univariate AUC is too low. A sensitivity sweep on this cutoff is on the further-work list.

Three anomalies worth flagging

Predictor	What we expected	What the data shows
`Param.BundleName` (AUC 74)	The action label itself, i.e. trivial.	It is the broader email context: *BookingConfirmation, InternetCheckIn, Pre_Departure_Outbound, AncillaryRecommender, Luggage, …* The *Luggage* bundle alone has positives-lift +261%, driving the high AUC. Legitimate signal, but near-tautological ("show luggage in luggage-themed emails").
`IsStaffStandBy` (AUC 66)	A staff flag, possibly a label leak.	**Almost no records are actually true** (52 out of 3.15M). The high AUC comes from the **MISSING vs false** split: MISSING records (37% of pop.) convert at −86% vs base, populated records (63%) at +50%. The flag is really a *data-completeness proxy*, not a staff indicator.
`BookingMonth` (AUC 72.5)	Calendar month, seasonality.	Bins read `<1.02`, `[1.02,2.02)`, `[2.02,3.02)`, `...`, `≥5.02`. Looks like *months until departure* (or since booking), not a calendar month. Pattern is U-shaped: extremes convert best. Semantics need confirmation.

All three need a one-question check with the Transavia/Pega team before any are accepted as causal signal.

The clearest business signal: message fatigue

`IH.Email.Out.Pending.pxLastGroupID`, bin lifts vs base rate:

Last-group bin	Lift	Interpretation
Promotions	**+125%**	cold customers respond best
ThirdParty	+40%	
FAQs	−11%	
BundleParent	−40%	
Luggage	**−50%**	recently shown luggage → **half** the CTR

Operational implication

A customer recently exposed to a luggage offer is roughly half as likely to engage with another one. Cadence / suppression rules tuned to this can raise effective CTR *without changing the model*.

Signals that could lift CTR but are currently inaccessible

Today the L5B15 model is mostly a *recency / fatigue machine*. Click-through prediction could likely be improved by stronger customer-level signals. Two complementary paths:

- **Cheaper**, predictors that are already configured in the model but stay inactive because the upstream data isn't flowing in. Fix the data pipeline, no feature-engineering needed.
- **Heavier**, customer or route-level features that would need to be designed (e.g. customer-segment clustering, route clusters, trip-purpose composites). Future-work track.

Predictor family (already configured)	Missing %	Status today
`Customer.CustomerKPIs.*CountTotal` (per-product history)	75–77%	inactive, would personalise
`IH.Push.Outbound.Received.*`	90%	inactive
`IH.Email.Outbound.Rejected.*`	87%	inactive
`IH.Email.Inbound.Rejected.*`	85%	inactive
`IH.Email.Inbound.Clicked.*`	93%	some active despite gap

Bridging back to the thesis (RQ3)

The two stories are the same story

The instability the thesis measures partly traces back to:

- predictors with very high missingness $\rightarrow$ unstable contributions across temporal splits (drives δ_d; KS = 0.36 on L5B15 temporal)
- NB leaning on a small set of correlated IH counters $\rightarrow$ DT rankings collapse on route splits (ρ as low as 0.51)

Fixing the upstream sourcing for the missing customer-history fields would reduce academic instability *and* could meaningfully increase CTR.

PART 3

Forward look: lifting luggage CTR

Three priority directions, anchored in the evidence

#	Proposed lever	Why now (evidence)
1	**Fix per-product purchase history data** (`CustomerKPIs.*`)	75%+ missing today; most personalisation features are blind. Unblocks a whole family of predictors.
2	**Cadence / suppression on recent luggage exposure**	Recent-luggage bin shows −50% lift; check current arbitration. Quick operational win, no model retrain.
3	**Validate the three predictor anomalies with the Transavia/Pega team**	`BundleName`, `IsStaffStandBy`, `BookingMonth`: all need a semantic check before they're trusted as causal signal.

Further directions for future investigation

Customer-segment clustering for personalisation | route-cluster features (beach / city / ski / business) | trip-purpose composite (duration × destination × children × weekday) | cross-channel fatigue score | sensitivity sweep on Pega's predictor-correlation cutoff.

Where the secondary work meets the primary work

Two reasons the CatBoost surrogate ($R^2 \approx 0.92$) outperforms the headline NB AUC of 76

1. NB cannot capture predictor *interactions*; the surrogate can.
2. NB's binning collapses high-cardinality categoricals; the surrogate handles them natively.

In other words: the surrogate is also a measurement of how much *predictive headroom* is left on the table by the current NB architecture.

Concrete next step Pega already supports

Pega ADM can run an **Adaptive Gradient Boosting (AGB)** model in *shadow mode* alongside the production NB, with no operational risk to live decisioning. This is a low-cost way to test whether the headroom the surrogate suggests actually translates into measurable AUC/CTR gains in Transavia's data. Worth piloting in parallel with the data-pipeline fixes.

PART 4

Next steps & discussion

Next steps

Thesis (next 4–6 weeks)

- Tighten the RQ3 numerical results across all four variants
- Sensitivity analysis: KS bin count, route-volume cutoff
- Write up Results chapter

Possible follow-ups (out of thesis scope)

- Validate the three predictor anomalies with the Transavia/Pega team
- Quantify the cadence/fatigue effect on a holdout
- Review `CustomerKPIs` data sourcing
- Pilot AGB in shadow mode alongside the production NB

Open questions for the room

1. Does `Param.BundleName` encode the action being scored, or the broader email context? How should we interpret its high AUC?
2. Is `IsStaffStandBy` populated for all bookings, or are missing values a known data gap?
3. What does `BookingMonth` actually measure: calendar month, months until departure, or something else?
4. Is there appetite to invest in the per-product purchase history pipeline (`CustomerKPIs`), or feature engineering to increase predictive power and CTR?

Thank you

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